Reflex Lab 01 Bill Of Materials

Required Parts

Qty	Part	Value / Type	Notes
1	ESP32 DevKit	ESP32-WROOM style	Course firmware targets ESP32 Arduino.
2	Mini breadboards	Solderless	Used beside the ESP32 to extend pin access.
1	Main breadboard	Solderless	Used for pot, LED, button, buzzer, and jumpers.
1	Potentiometer	10 kOhm	Analog input to GPIO34.
1	LED	Standard red or similar	Output indicator.
1	Resistor	330 ohm	LED current limiting.
1	Passive piezo buzzer	Passive piezo disc/buzzer	GPIO26 output path.
1	Resistor	1 kOhm, brown-black-black-brown-brown	Buzzer series resistor.
1	Tactile button	2-pin or 4-pin breadboard button	GPIO27 to GND, using ESP32 internal pullup.
several	Jumper wires	Male-male	Use colors consistently where possible.
1	USB cable	Data-capable	Needed for programming and live serial capture.

Acceptable Substitutions

- LED color can change if the resistor remains in series.
- A 4-pin tactile switch is fine; use one pair that closes when pressed.
- The buzzer must be passive for tone/PWM-style behavior. An active buzzer may work differently and should be documented as a substitution.

Safety Notes

- Do not connect the potentiometer to 5V for this course.
- Do not drive the LED without a resistor.
- Do not connect motors, relays, solenoids, or external high-current loads.

- Unplug USB before changing the 01B button/buzzer wiring.